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# Ultra-early prediction of lithium-ion battery cycle life based on assembled capacity curve extracted from a single cycle

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## HIGHLIGHTS

- Proposes an ultra-early lithium-ion battery cycle life prediction framework.
- Defines the assembled capacity curve based on charging/discharging capacities.
- Demonstrates the framework effectiveness using three open battery datasets.

## ARTICLE INFO

*Keywords:*
Lithium-ion battery
Assembled capacity curve
Ultra-early cycle life prediction
Single cycle
Convolutional neural network

## ABSTRACT

This paper proposes a lithium-ion battery cycle life prediction framework utilizing the so-called "assembled capacity curve" extracted from a single cycle during the ultra-early stage (e.g., the second cycle) of the battery cycling test. The assembled capacity is determined as the difference between the charging capacity and the discharging capacity of the same cycle. The proposed framework employs AlexNet to extract the features of the image constructed based on the assembled capacity curve. The proposed framework is evaluated using three battery datasets. Utilizing the assembled capacity curve extracted from the second cycle, the battery cycle life prediction errors are 11.36%, 15.27%, and 7.86% for the three datasets in terms of the mean absolute percentage error (MAPE), respectively. When other early cycles (e.g., the fifth and tenth cycles) are utilized to build the assembled capacity curves, the prediction errors are at the same level. Moreover, the proposed framework outperforms three feature-based methods (i.e., ElasticNet, LightGBM, and XGBoost) and another three feature-free methods (i.e., ResNet18, SqueezeNet, and 1D-CNN) in reducing the battery cycle life prediction error. All these evaluation results demonstrate the effectiveness of the proposed framework.

## 1. Introduction

As a popular electrochemical energy storage technology, lithium-ion batteries feature a high energy density, a low self-discharge rate, and a long cycle life. Therefore, they have been widely employed in both low-power systems such as consumer electronics [1,2] and high-power systems such as electric vehicles [3,4] and energy storage stations coupled with electric power grids [5,6]. However, battery aging and degradation result in compromised performance and increased safety concerns [7]. Therefore, the battery state of health (SOH) needs to be accurately estimated and its remaining useful life (RUL) needs to be precisely predicted to enhance the safety and economy of battery-based energy storage systems [8–10]. In particular, it is of great interest to predict the battery RUL during an early stage to boost the battery economy during its entire cycle life [11–14]. In the meantime, if a rapid prediction of the battery RUL can be implemented using the information extracted from only one or at most a few charging–discharging cycles, a quick evaluation of the battery can be performed, which can be extremely beneficial to new battery development and used battery recycling [15,16]. Therefore, this paper considers the problem of predicting the battery cycle life using the information extracted from a single charging–discharging cycle during an ultra-early stage of the battery cycling test. In particular, for most batteries in the Severson [8], Zhu [9], and Wang [10] datasets examined in this paper, the earliest cycle with usable data is the second cycle. Therefore, this paper mainly utilizes the second cycle to illustrate the proposed ultra-early battery cycle life prediction framework.

To estimate the battery SOH and predict its RUL, numerous approaches have been proposed, which fall into two main categories: model-based methods and data-driven methods [17–23]. Model-based methods utilize electrochemical models (EMs) or equivalent circuit models (ECMs) of the battery. EMs such as the pseudo-two-dimension

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(P2D) model establish mathematical models for critical physical processes within the battery, which can describe the behaviors of various particles during the charging or discharging processes [24,25]. To develop an accurate EM, numerous parameters may need to be identified. In the meantime, solving multiple partial differential equations can be computationally expensive. To reduce the complexity of EMs, ECMs typically utilize electrical components such as capacitors and resistors to characterize the external volt–ampere behavior of the battery [24,26].

In the category of data-driven methods, various frameworks have been proposed for SOH estimation and RUL prediction [9,27–32]. In particular, for battery RUL prediction, data-driven methods can be further categorized as time series forecasting (TSF) methods, feature-based methods, and feature-free methods. For TSF methods, the historical capacity degradation trajectories are used as the model inputs. To predict the future trajectory, popular TSF tools including recurrent neural network (RNN), long short-term memory (LSTM), gated recurrent unit (GRU), and Gaussian process regression (GPR) have been examined [33–39]. TSF methods usually need extensive experiments to collect sufficient data [40]. To predict the future trajectory, TSF methods typically require 40%–70% of the data during the battery cycle life [41], which makes them not the best option for ultra-early prediction of the battery cycle life because only a limited amount of data is available during the ultra-early stage.

To predict the battery RUL using feature-based methods, machine learning or deep learning models are usually trained using the features extracted from the battery physical parameters such as the voltage, current, and temperature [42–46]. For instance, an ElasticNet model is trained using the features extracted from the $Q(V)$ (i.e., the relationship between the battery capacity and its voltage) curves of the first 100 cycles [8]. Other battery physical parameters that can be used for feature extraction include the discharging capacity, discharging current, terminal voltage, temperature, and electrochemical impedance [47–50]. Although feature-based methods can realize rapid predictions of the battery cycle life and they can be deployed in real-world applications because of their relatively small model sizes, they usually require expert knowledge to extract effective features. However, it may be challenging to extract such effective features during the ultra-early stage of the battery operation because the degradation pattern may be unclear [51].

Compared to feature-based methods, feature-free methods remove the dependence on handcraft features. In fact, various convolutional neural networks (CNNs) are explored to predict the battery cycle life due to their capabilities of parameter sharing and automatic feature extraction [52–54]. For example, a CNN model is trained to automatically extract features from the images constructed using the $Q(V)$ curves of the first 100 discharging cycles [53]. To construct images of this kind, other feature curves such as the incremental capacity curve and the capacity difference curve have also been utilized [55]. Utilizing such feature curves can narrow down the solution space and enhance the generalization capability of the CNN model. In addition to exploring such feature curves, some CNN-based methods have also been developed to predict the battery RUL during the ultra-early stage. For example, a hybrid method leveraging CNN and LSTM is proposed to predict the battery RUL based on the data of the first 30 cycles [56]. Furthermore, multiple methods have been proposed to utilize the data extracted from a single cycle. For instance, a one-dimensional CNN (1D-CNN) model is proposed to predict the battery capacity degradation trajectory, which is based on the voltage, current, and temperature extracted from a single cycle [52]. As another example, to predict the $Q(V)$ curves of the future cycles, a sequence-to-sequence (Seq2Seq) model is developed based on the $Q(V)$ curve of a single cycle [57]. Moreover, an ultra-early battery cycle life prediction framework based on the visualized data (i.e., voltage, current, and capacity) extracted from a single cycle is developed [58].

While three battery physical parameters (i.e., voltage, current, and capacity) are utilized in [58], this paper focuses on the capacity behavior and utilizes the so-called "assembled capacity curve" extracted from a single cycle to develop an ultra-early battery cycle life prediction framework. Specifically, the assembled capacity is determined as the difference between the charging capacity and the discharging capacity of the same cycle, which carries rich information such as the charging and discharging capacities as well as the charging and discharging rates. This paper makes the following contributions:

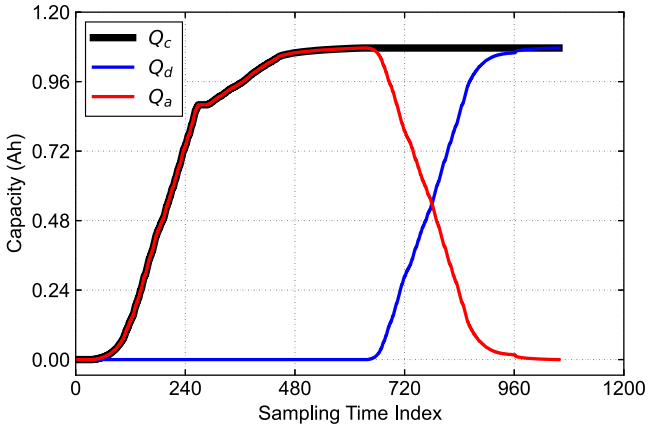



**Fig. 1.** Illustration of assembled capacity curve.

- First, an ultra-early battery cycle life prediction framework based on the assembled capacity curve is proposed. Using the assembled capacity curve extracted from the second cycle, the battery cycle life prediction errors are 11.36%, 15.27%, and 7.86% for the Severson [8], Zhu [9], and Wang [10] datasets in terms of the mean absolute percentage error (MAPE), respectively. Note that in addition to the Severson dataset examined in [58], this paper examines two more datasets (i.e., the Zhu and Wang datasets). Since these datasets cover batteries of different chemistries (i.e., LFP in the Severson dataset, NCA and NCM in the Zhu dataset, and NCM in the Wang dataset), this paper demonstrates the general applicability of the proposed framework.
- Second, while [58] only utilizes the second cycle, this paper utilizes the first cycle and several later cycles (e.g., the fifth and tenth cycles) to build the assembled capacity curves in addition to the second cycle. These evaluation results show that the battery cycle life prediction errors using different cycles are at the same level. Moreover, the inverse assembled capacity curve is extracted from the second cycle, which also leads to approximately equal prediction errors. Therefore, a more comprehensive investigation of the information extracted from a single cycle is conducted in this paper compared to [58].
- Finally, the proposed framework is evaluated against three feature-based methods (i.e., ElasticNet, LightGBM, and XGBoost) and another three feature-free methods (i.e., ResNet18, SqueezeNet, and 1D-CNN). All the evaluation results demonstrate the effectiveness of the proposed framework in predicting the battery cycle life during an ultra-early stage.

The remainder of this paper is organized as follows. Section 2 illustrates the assembled capacity curve. Section 3 presents the ultra-early battery cycle life prediction framework. Section 4 evaluates the proposed framework. Section 5 concludes this paper.

## 2. Assembled capacity curve

This paper proposes to predict the battery cycle life based on the assembled capacity curve extracted from an ultra-early cycle of the battery cycling test, as illustrated in Fig. 1. The battery under test is the "b1c5" cell in batch 1 in the Severson dataset [8], which is of

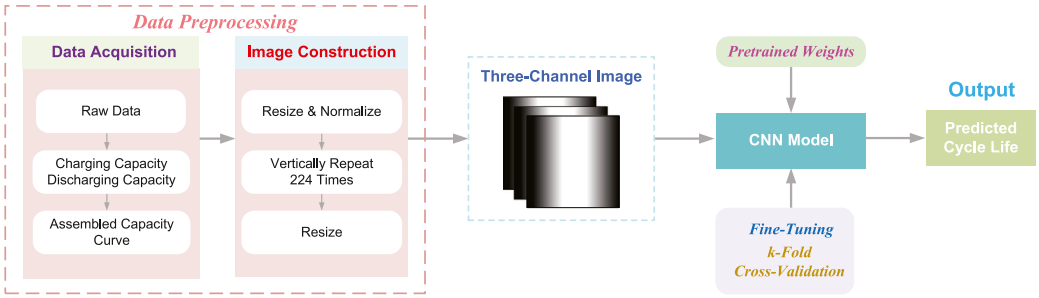



**Fig. 2.** Ultra-early battery cycle life prediction framework.

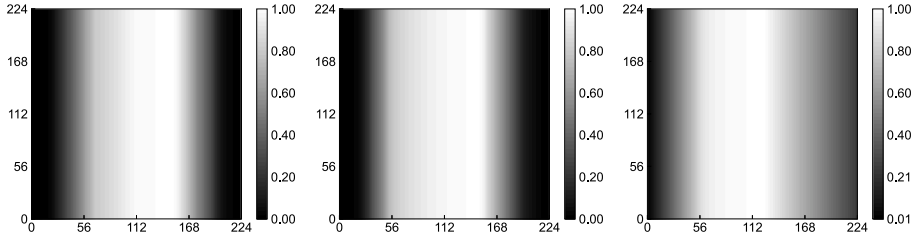



**Fig. 3.** Illustration of assembled capacity images of three datasets. (a) Severson dataset. (b) Zhu dataset. (c) Wang dataset.

the LFP chemistry with a nominal capacity of 1.1 Ah. For the cells in batch 1, the second cycle is the earliest cycle with usable test data because the sampling rate for the first cycle is too high and the test data of the first cycle is excluded from the dataset. For consistency, this paper mainly focuses on the assembled capacity curve extracted from the second cycle for the cells in the three datasets [8–10] although the first cycle data may also be usable for certain cells, as elaborated in Section 4.4. The test procedure for the battery and the construction flow of the assembled capacity curve are elaborated as follows.

The battery is charged from the sampling time index 1 to index 627. Since index 628, the battery is discharged until index 1058. The battery is at rest until index 1059. The charging capacity is denoted as $Q_c$, which increases from 0 to 1.0761 Ah when the index increases from 1 to 626 and remains at 1.0761 Ah until index 1059 (i.e., the end of the second cycle). The discharging capacity is denoted as $Q_d$, which is 0 from index 1 to index 627. Then, $Q_d$ increases from 4.6147e−06 to 1.0761 Ah when the index increases from 628 to 1053. From index 1054 to index 1059, $Q_d$ is fixed at 1.0761 Ah. The assembled capacity is calculated as $Q_a = Q_c - Q_d$ for the entire cycle beginning with index 1 and ending with index 1059. Both the $Q_c$ and $Q_d$ curves carry rich information such as the charging and discharging capacities as well as the charging and discharging rates. The $Q_a$ curve retains such information and adds more information such as the difference between the charging and discharging capacities. Therefore, this paper explores the feasibility of utilizing the assembled capacity curve to predict the battery cycle life during the ultra-early stage.

## 3. Ultra-early battery cycle life prediction framework

### 3.1. Overview

The proposed ultra-early battery cycle life prediction framework is illustrated in Fig. 2. In the data acquisition phase, the charging capacity and the discharging capacity are extracted from a single cycle (e.g., the second cycle) during the ultra-early stage. Then, the assembled capacity curve is obtained using the procedure elaborated in Section 2. Next, the assembled capacity curve is transformed into a three-channel assembled capacity image, as elaborated in Section 3.2. Finally, the assembled capacity image and the cycle life label (i.e., the index of the cycle when the battery capacity decreases to 80% of the nominal capacity) are utilized to train a CNN model for ultra-early prediction of the battery cycle life, as elaborated in Section 4.1.

### 3.2. Construction of assembled capacity image

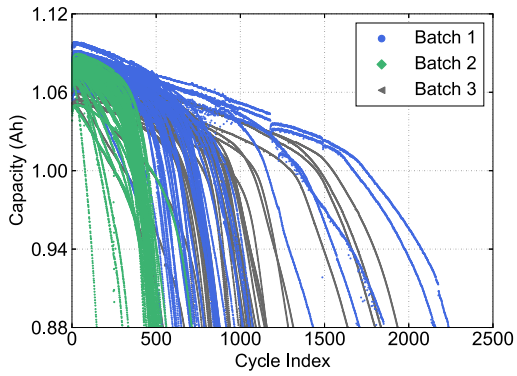
The assembled capacity image is constructed following the flow illustrated in Fig. 2. The $Q_a$ curve can be seen as a one-dimensional (1D) vector. To construct a two-dimensional (2D) image, the following steps are taken. First, the length $L$ of the $Q_a$ vector is resized into 1000. Second, the resized $Q_a$ vector is normalized into the range of [0, 1]:

$$z_{\text{norm}} = \frac{z - z_{\min}}{z_{\max} - z_{\min}} \tag{1}$$

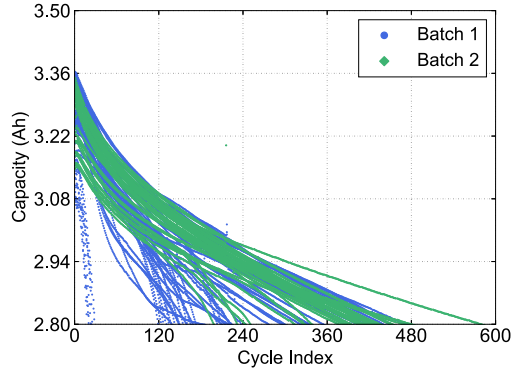
where $z$ is the raw data, $z_{\text{norm}}$ is the normalized data, $z_{\min}$ and $z_{\max}$ are the minimum and maximum values of the raw data, respectively. Third, the normalized $Q_a$ vector is vertically repeated 224 times. The image size is then 224 × 1000. To meet the input size requirement of the CNN model, the $Q_a$ image is resized to 224 × 224. For instance, Fig. 3 shows three samples of the $Q_a$ image: one for each dataset. The value of each pixel in the $Q_a$ image is mapped to a gray level: the whitest level represents the maximum value of the assembled capacity while the blackest level represents the minimum value. Note that the minimum value of the $Q_a$ image shown in Fig. 3(c) is 0.01 instead of 0.00 because of the resize process. Finally, each $Q_a$ image is repeated three times to construct a three-channel image, which is eventually fed into the CNN model.

### 3.3. CNN model architecture

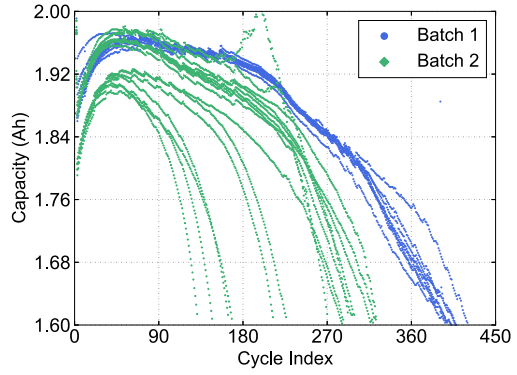
The architecture of the CNN model utilized in the proposed framework is AlexNet [59]. In the fields of deep learning and computer vision, AlexNet represents a CNN architecture milestone, which comprises five convolutional layers, three fully connected layers, rectified linear unit (ReLU) activation functions, max-pooling layers, and dropout regularization. Therefore, AlexNet is capable of automatically learning features from a vast array of images without introducing manual feature engineering. To meet the needs of the regression task, the 1000 output features of the original AlexNet are replaced with a single output feature.




(a)




(b)




(c)

**Fig. 4.** Battery capacity degradation trajectories of three datasets. (a) Severson dataset. (b) Zhu dataset. (c) Wang dataset.

## 4. Evaluation of proposed framework

### 4.1. Evaluation setup

#### 4.1.1. Battery datasets

This paper utilizes three battery datasets [8–10] to evaluate the proposed framework. The key characteristics of them are elaborated as follows.

The Severson dataset [8] contains 124 LFP cells divided into three batches, each with 41, 43, and 40 cells, respectively. These cells are rated with a nominal capacity of 1.1 Ah. All the 124 cells are utilized to evaluate the proposed framework. Fig. 4(a) shows that the cycle life of the 124 cells varies between 148 and 2237 cycles. Considering that the cycle life distribution of the 124 batteries is unbalanced, this dataset is partitioned as follows: (1) Out of the 11 batteries with a relatively long cycle life (i.e., no less than 1200 cycles), one battery is randomly selected and included in the test set. (2) All the three batteries with a relatively short cycle life (i.e., less than 400 cycles) are excluded from the test set. (3) 25% (27.5, rounded to 28) of the remaining 110 batteries are randomly selected and included in the test set. In total, there are 29 batteries in the test set. The remaining 95 batteries form the training and validation set.

**Table 1**
Summary of three battery datasets.

| Dataset | Severson | Zhu | Wang |
|---|---|---|---|
| Manufacturer/ Model | A123 systems/ APR18650M1A | LG/INR18650-35E<br>Samsung/INR18650-MJ1<br>Samsung/INR18650-25R | LISHEN/– |
| Positive electrode material | LFP | Batch 1: NCA;<br>Batch 2: NCM;<br>Batch 3: NCM+NCA. | NCM |
| Temperature (°C) | 30 | Batches 1 and 2: 25, 35, 45;<br>Batch 3: 25. | Room temperature |
| Number of charging–discharging policies | 69 | 11 | 6 |
| Number of cells | 124 | 130 | 55 |
| Nominal capacity (Ah) | 1.1 | Batches 1 and 2: 3.5;<br>Batch 3: 2.5. | 2.0 |
| Form factor | 18650 | 18650 | 18650 |

• Specific positive electrode materials are elaborated in the main text.

The Zhu dataset [9] also consists of three batches: batch 1 with 66 NCA cells, batch 2 with 55 NCM cells, and batch 3 with 9 NCM+NCA cells. In total, 130 cells are included in this dataset. The cells in batches 1 and 2 are rated with a nominal capacity of 3.5 Ah and the batch 3 cells are rated at 2.5 Ah. Since too few cells are in batch 3, only the cells in batches 1 and 2 are used to evaluate the proposed framework. Moreover, the cells that do not reach 80% of their nominal capacity after the cycling tests are excluded. In addition, the cell "NCA_CY25-025_1-#1" is considered an outlier and removed because of its significantly different $Q_a$ curve. Finally, 90 cells are utilized, of which 55 cells are in batch 1 and 35 cells are in batch 2. The cycle life of the 90 cells varies between 18 and 580 cycles, as shown in Fig. 4(b). For dataset partitioning, 27% (24.3, rounded to 25) of the 90 cells are randomly selected and included in the test set. The remaining 65 cells form the training and validation set.

The Wang dataset [10] includes 55 NCM cells divided into six batches and each batch includes 8, 15, 8, 8, 8, and 8 cells, respectively. All the cells are rated with a nominal capacity of 2.0 Ah. The cells in batches 1 and 2 are tested using fixed charging–discharging strategies while the cells in batches 3–6 are tested using random charging–discharging strategies. Since the cycle life is associated with the charging–discharging strategy, this paper only utilizes the 23 cells in batches 1 and 2 and drops the cells in batches 3–6. As shown in Fig. 4(c), the cycle life of the 23 cells varies between 131 and 420 cycles. 34% (7.82, rounded to 8) of the 23 cells are randomly chosen to form the test set. The remaining 15 cells form the training and validation set.

A summary of the three battery datasets is listed in Table 1. These batteries are of at least five models (the Wang dataset only specifies the manufacturer, but not the battery model(s)), which are made by four manufacturers. In total, five different positive electrode materials are used in these batteries: LFP for the Severson dataset, NCA, NCM, and NCM+NCA for the Zhu dataset, and a different NCM material for the Wang dataset, of which four materials are used by the cells in the evaluation (i.e., the NCM+NCA material is excluded). Specifically, all the cells in the Severson dataset are of the LFP chemistry. For the Zhu dataset, the NCA and NCM materials refer to $LiNi_{0.86}Co_{0.11}Al_{0.03}O_2$ and

**Table 2**
Evaluation results using assembled capacity curve extracted from second cycle.

| | Severson dataset | | Zhu dataset | | Wang dataset | |
|---|---|---|---|---|---|---|
| | RMSE (cycles) | MAPE (%) | RMSE (cycles) | MAPE (%) | RMSE (cycles) | MAPE (%) |
| Exp. 1 | 100.97 | 8.93 | 54.83 | 14.48 | 27.29 | 9.56 |
| Exp. 2 | 131.43 | 12.35 | 58.82 | 16.71 | 26.21 | 10.76 |
| Exp. 3 | 130.34 | 12.82 | 47.26 | 12.97 | 13.74 | 5.06 |
| Exp. 4 | 121.35 | 11.33 | 48.24 | 16.91 | 28.93 | 6.06 |
| Average | 121.02 | 11.36 | 52.29 | 15.27 | 24.04 | 7.86 |

$LiNi_{0.83}Co_{0.11}Mn_{0.07}O_2$, respectively, while NCM+NCA is made of 42 (3) wt.% $Li(NiCoMn)O_2$ blended with 58 (3) wt.% $Li(NiCoAl)O_2$. For the Wang dataset, the specific NCM material is $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$. These cells are cycled at different temperatures. In fact, the cells in the Severson dataset are tested at a constant temperature of 30 °C. As for the cells in the Zhu dataset, the cells in batches 1 and 2 are tested at three different temperatures: 25, 35, and 45 °C while the cells in batch 3 are cycled at 25 °C. The cells in the Wang dataset are cycled at the room temperature. The cycling policy also varies from dataset to dataset. The number of different charging–discharging policies in the Severson, Zhu, and Wang datasets is 69, 11, and 6, respectively. Although the form factors of the cells in the three datasets are all 18650, the nominal capacities of these cells vary. In total, four different capacities are covered in the three datasets: 1.1, 3.5, 2.5, and 2.0 Ah, of which three capacities are examined in the evaluation (i.e., 2.5 Ah is excluded). Although the three battery datasets examined in this paper are not an exhaustive list of all the publicly available datasets, they do cover a relatively wide range of commercially available lithium-ion batteries. Therefore, the general applicability and effectiveness of the proposed framework can be demonstrated using these three battery datasets.

#### 4.1.2. CNN model training

To train the CNN model (i.e., AlexNet) used in the proposed framework, a fine-tuning strategy is adopted because the sizes of the three battery datasets are relatively small [55]. Specifically, the fine-tuning strategy uses a pretrained model that is trained on a relatively large dataset (might not be a battery dataset) and has a strong capability of extracting effective features. When the pretrained model is trained on a new dataset, certain layers of it will be frozen and the parameters (weights) of these frozen layers will not be updated during the training process. The pretrained AlexNet is obtained from torchvision [60], which is a PyTorch library that provides popular pretrained models together with their pretrained weights. In this paper, all the AlexNet layers are not frozen and they are trained directly based on the pretrained weights.

The $k$-fold cross-validation method is utilized for model selection in this paper. For different datasets, the specific cross-validation method varies, as elaborated in Section 4.2. The $k$-fold cross-validation method makes a full use of the available data and avoids model overfitting. In particular, the model with the best performance across all the $k$ trials is selected and applied to the test set. To be clear, the entire process of dataset partitioning, model selection, and model testing is referred to as an experiment (denoted as Exp.) in this paper. For each dataset, a set of four experiments is performed to avoid contingencies.

The mean square error (MSE) is selected as the loss function, which is determined as follows:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

where $n$ is the number of batteries, $y_i$ is the observed cycle life of battery $i$, and $\hat{y}_i$ is the predicted cycle life of battery $i$.

#### 4.1.3. Performance evaluation metrics

The performance of the proposed framework is evaluated using two metrics: root mean square error (RMSE) and mean absolute percentage error (MAPE), which are determined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

$$\mathrm{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \tag{4}$$

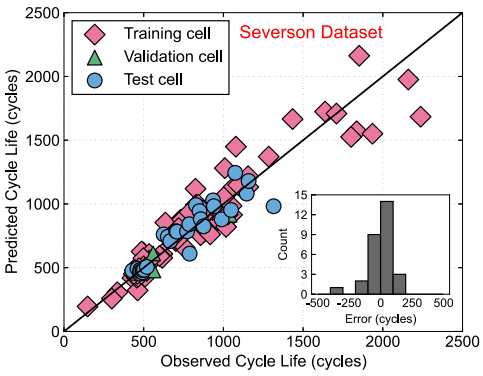
### 4.2. Evaluation results using assembled capacity curve extracted from second cycle

This section evaluates the proposed framework using the assembled capacity curve extracted from the second cycle. For model selection, the Severson, Zhu, and Wang datasets utilize the 19-fold, 15-fold, and 15-fold cross-validation methods, respectively. The evaluation results are listed in Table 2. In general, the proposed framework is effective in that the battery cycle life prediction errors are acceptably low. For all the three datasets, the average MAPEs of the test sets are less than 16%: the highest of 15.27% for the Zhu dataset, the lowest of 7.86% for the Wang dataset, and 11.36% for the Severson dataset, respectively. On the other hand, the average RMSEs are 121.02, 52.29, and 24.04 cycles for the Severson, Zhu, and Wang datasets, respectively.

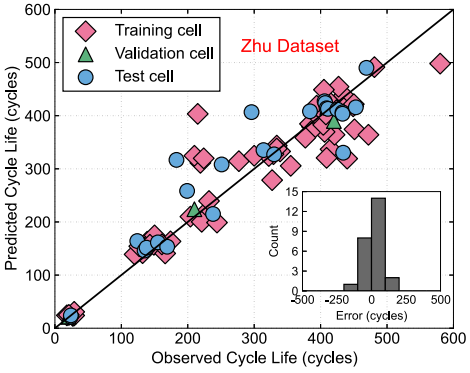
Fig. 5 shows the visualization results for the individual experiments with the lowest MAPEs: Exp. 1 for the Severson dataset, Exp. 3 for the Zhu dataset, and Exp. 3 for the Wang dataset, respectively. The insets in Fig. 5 show the histograms of the prediction errors, which are calculated as the differences between the predicted cycle life and the observed cycle life. In general, the proposed framework performs well in all the three datasets. Graphically, for all the three datasets, most of the blue dots representing the batteries in the test sets are close to the diagonal. In the meantime, the highest and the second highest bars in the histograms suggest that the errors for a large portion of the batteries in the test sets are relatively low. As for the maximum error (in terms of the absolute value), the proposed framework results in 333.13, 133.81, and 31.16 cycles for the Severson, Zhu, and Wang datasets, respectively.

### 4.3. Evaluation results using inverse assembled capacity curve extracted from second cycle

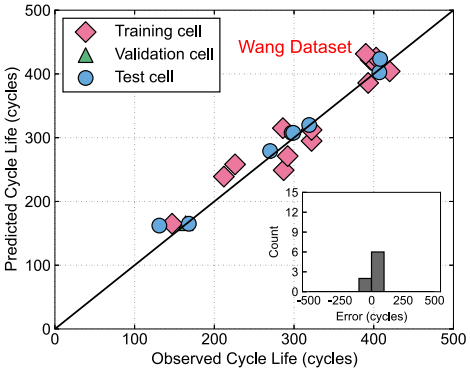
This section evaluates the proposed framework using the inverse assembled capacity curve extracted from the second cycle. In particular, the inverse assembled capacity is determined as $Q_{ia} = Q_d - Q_c$, which is the inverse of the assembled capacity $Q_a = Q_c - Q_d$ examined in Section 4.2. The evaluation results are listed in Table 3. Note that when the experiment index is the same in Tables 2 and 3, the dataset partition is also the same. In terms of both RMSE and MAPE, the results in Table 3 are similar to those in Table 2. For instance, the average MAPEs are 11.49% versus 11.36% for the Severson dataset, 14.05% versus 15.27% for the Zhu dataset, and 7.31% versus 7.86% for the Wang dataset, respectively. As for the minimum MAPE, Table 3 shows that for the Severson, Zhu, and Wang datasets, Exp. 3, Exp. 3, and Exp. 4 lead to the minimum MAPEs while the experiments associated with the minimum MAPEs are Exp. 1, Exp. 3, and Exp. 3 in Table 2. For RMSE, similar




(a)




(b)




(c)

**Fig. 5.** Battery cycle life prediction results of three datasets. (a) Severson dataset. (b) Zhu dataset. (c) Wang dataset.

observations can be made. Despite such specific differences between the results in Tables 2 and 3, Sections 4.2 and 4.3 show that the proposed framework is effective in predicting the battery cycle life using either the original or the inverse assembled capacity curves extracted from the second cycle.

### 4.4. Evaluation results using assembled capacity curve extracted from first cycle

As mentioned in Section 2, the second cycle is the earliest cycle with usable data for certain batteries in the Severson dataset. Therefore, Sections 4.2 and 4.3 evaluate the proposed framework using the data of the second cycle for all the three datasets. This section examines the feasibility of using the first-cycle data to predict the cycle life for the batteries with usable data in the very first cycle. For the Wang dataset, although the first-cycle data is available, the first cycle is only used as a test drive: all the batteries are tested using exactly the same procedure, which is not the charging–discharging policy to be examined. Therefore, the Wang dataset is not utilized in this section. For the Severson dataset, the first-cycle data for batch 1 is unusable. Therefore, this section only considers the 83 batteries in batches 2 and 3 with usable first-cycle data. For the Zhu dataset, the first-cycle data for 86 batteries is usable and these batteries are examined in this section. In terms of dataset partition, 19 batteries out of the 83 batteries in the Severson dataset form the test set. The remaining 64 batteries form the training and validation set. For the Zhu dataset, 22 batteries form the test set and the remaining 64 batteries form the training and validation set. For model selection, both the Severson and Zhu datasets utilize the 16-fold cross-validation method. Note that this section builds the assembled capacity curve using $Q_a = Q_c - Q_d$, which is the same as the one used in Section 4.2.

The evaluation results are listed in Table 4. Compared to the evaluation results obtained using the second-cycle data listed in Table 2, the prediction errors are at the same level in terms of both RMSE and MAPE. In fact, the prediction errors in Table 4 are slightly lower. For instance, the average MAPEs for the Severson dataset are 9.72% and 11.36% in Tables 4 and 2, respectively. Therefore, this section demonstrates that the proposed framework is effective in predicting the battery cycle life using the first-cycle data.

### 4.5. Evaluation results using assembled capacity curves extracted from later cycles

In addition to the second and first cycles examined in Sections 4.2 and 4.4, respectively, some relatively late cycles compared to them are examined in this section to more comprehensively evaluate the proposed framework. Again, all the three battery datasets are utilized and the assembled capacity curves are built using $Q_a = Q_c - Q_d$. For the Severson and Wang datasets, cycles #10, #50, and #100 are examined in this evaluation. For the Zhu dataset, the shortest cycle life of the cells is 18 cycles. Therefore, to ensure that the same amounts of batteries are used in this evaluation, cycles #5, #10, and #15 are examined, which are different from those for the Severson and Wang datasets. Note that the dataset partition and model selection are exactly the same as those in Section 4.2. The evaluation results are shown in Fig. 6. For comparisons, the results for the second cycle are also shown. In general, two observations can be made.

First, for all the three battery datasets, the performance of the proposed framework is at the same level when different cycles are used. Take the Severson dataset for instance. Fig. 6(a) shows that in terms of the average RMSE, cycles #2, #10, #50, and #100 result in 121.02, 123.73, 134.17, and 140.82 cycles, respectively. Clearly, cycles #2 and #100 lead to the lowest (121.02 cycles) and highest (140.82 cycles) average RMSEs, respectively. With respect to 121.02, 140.82 is 19.8 higher, or equivalently, 16.4% higher in percentage. Following this approach, the difference (in terms of percentage with respect to the lowest error) between the highest and lowest errors is calculated for the results shown in Figs. 6(b) through 6(f). In terms of percentage, the maximum difference between the highest and lowest errors is 100.9%, which is observed in Fig. 6(f) while the minimum difference of 16.4% is observed in Fig. 6(a).

Second, the cycle leading to the minimum error varies. For Figs. 6(a) through 6(f), the minimum errors are obtained using cycles #2, #10, #15, #15, #2, and #2, respectively. While it seems that an earlier cycle results in a smaller error for the Severson (cycles #2 and #10) and Wang (cycle #2) datasets, the latest cycle (cycle #15) leads to a smaller error for the Zhu dataset. Therefore, additional work needs to be carried out to more rigorously examine this observation. In summary, this section shows that the prediction errors are at the same level in terms of both the RMSE and the MAPE when using the data extracted from different cycles.

**Table 3**
Evaluation results using inverse assembled capacity curve extracted from second cycle.

| | Severson dataset | | Zhu dataset | | Wang dataset | |
|---|---|---|---|---|---|---|
| | RMSE (cycles) | MAPE (%) | RMSE (cycles) | MAPE (%) | RMSE (cycles) | MAPE (%) |
| Exp. 1 | 138.40 | 11.12 | 60.63 | 14.14 | 28.75 | 9.97 |
| Exp. 2 | 160.86 | 13.24 | 58.83 | 17.48 | 27.11 | 10.98 |
| Exp. 3 | 100.89 | 9.87 | 43.74 | 11.68 | 12.90 | 4.77 |
| Exp. 4 | 151.46 | 11.73 | 42.10 | 12.90 | 13.67 | 3.50 |
| Average | 137.90 | 11.49 | 51.33 | 14.05 | 20.61 | 7.31 |

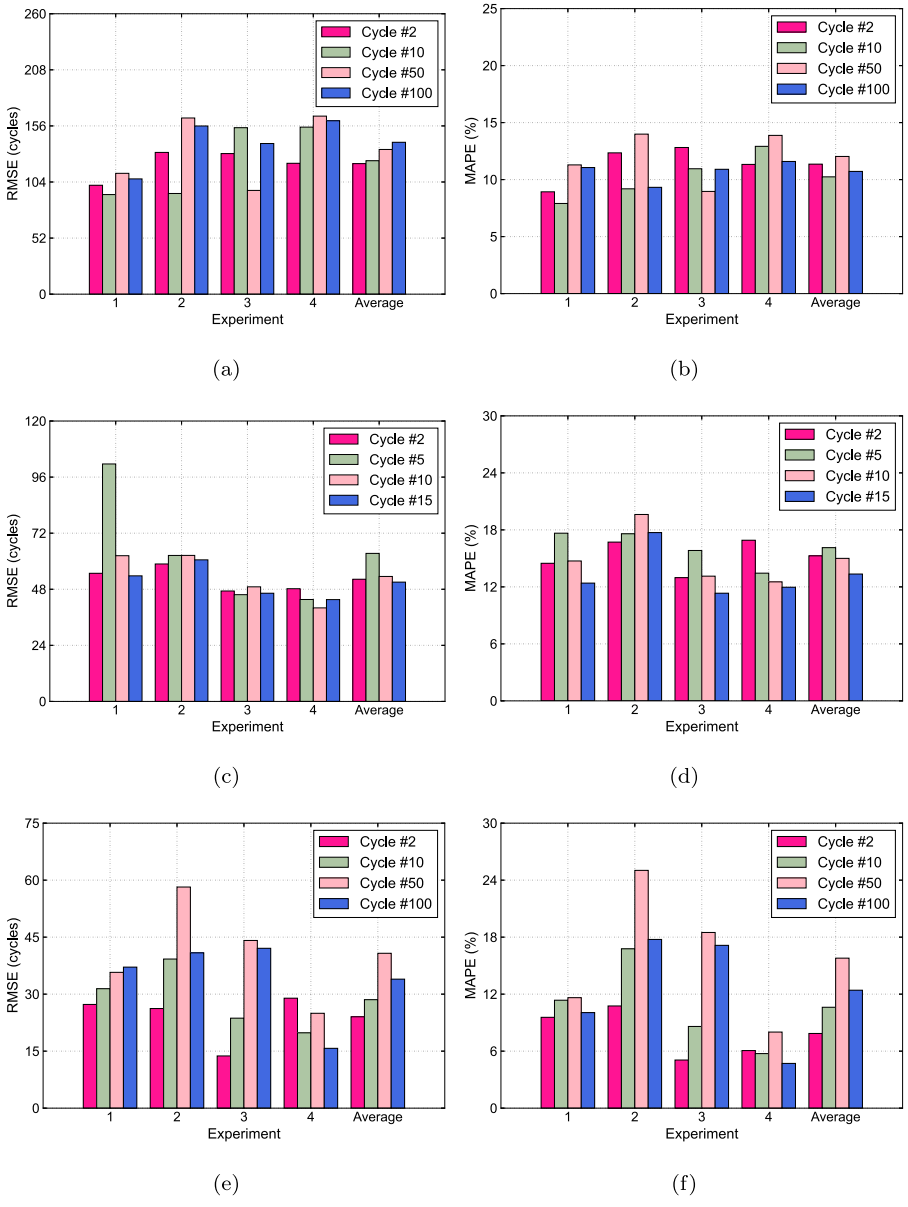



**Fig. 6.** Effects of different cycles on average RMSE and average MAPE. (a) Severson dataset: RMSE. (b) Severson dataset: MAPE. (c) Zhu dataset: RMSE. (d) Zhu dataset: MAPE. (e) Wang dataset: RMSE. (f) Wang dataset: MAPE.

### *4.6. Evaluation results: Comparison with feature-based methods*

To demonstrate the advantages of the proposed framework, a comparison is conducted between the proposed framework and three feature-based methods: ElasticNet, light gradient boosting machine (LightGBM), and extreme gradient boosting (XGBoost). To ensure a fair comparison, the three feature-based methods are also implemented using the evaluation setup illustrated in Section 4.1. All the three battery datasets are utilized in this comparison. For the three feature-based methods, the following five statistical features are extracted from the assembled capacity curve: maximum, mean, variance, skewness, and kurtosis. Note that the minimum values of the assembled capacity curves are almost zero and little information can be extracted from them. Therefore, the minimum value is not used as a feature. This section also builds the assembled capacity curve for the second cycle using $Q_a = Q_c - Q_d$.

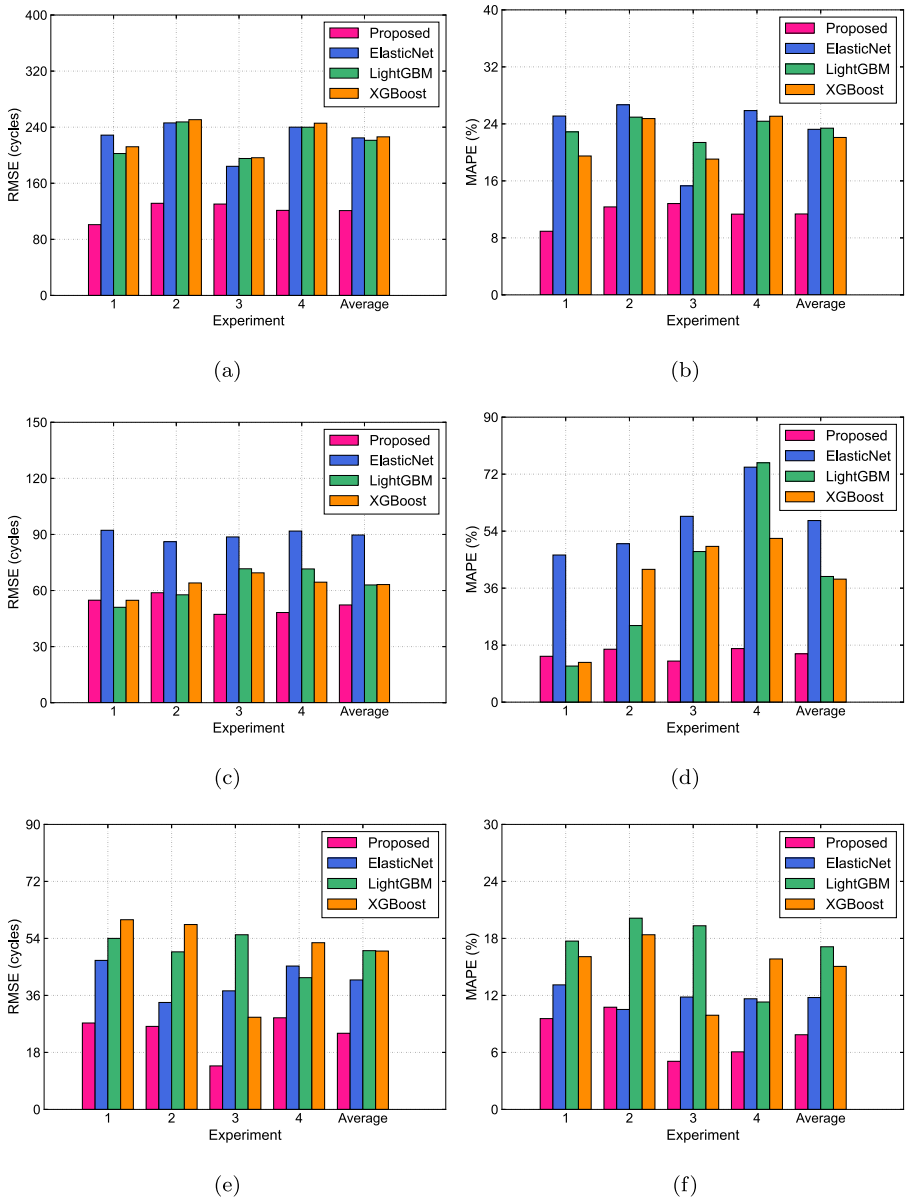



**Fig. 7.** Performance comparison between proposed framework and feature-based methods. (a) Severson dataset: RMSE. (b) Severson dataset: MAPE. (c) Zhu dataset: RMSE. (d) Zhu dataset: MAPE. (e) Wang dataset: RMSE. (f) Wang dataset: MAPE.

**Table 4**
Evaluation results using assembled capacity curve extracted from first cycle.

| | Severson Dataset | | Zhu Dataset | |
|---|---|---|---|---|
| | RMSE (cycles) | MAPE (%) | RMSE (cycles) | MAPE (%) |
| Exp. 1 | 78.44 | 7.91 | 35.09 | 8.01 |
| Exp. 2 | 127.62 | 12.75 | 41.96 | 18.03 |
| Exp. 3 | 117.87 | 10.25 | 64.42 | 16.20 |
| Exp. 4 | 73.24 | 7.96 | 57.94 | 13.04 |
| Average | 99.29 | 9.72 | 49.85 | 13.82 |

The comparison results for the three datasets are shown in Fig. 7. A general observation is that for all the three datasets, the proposed framework outperforms all the three feature-based methods in terms of both the average RMSE and the average MAPE. Take the Severson dataset as an example. Fig. 7(a) shows that in terms of the average RMSE, the proposed framework, ElasticNet, LightGBM, and XGBoost lead to 121.02, 224.75, 221.31, and 226.24 cycles, respectively. Out of the three feature-based methods, LightGBM results in the lowest average RMSE of 221.31 cycles, which is 1.83 times of the average RMSE associated with the proposed framework (i.e., 121.02 cycles). Similarly, Fig. 7(b) shows that the average MAPEs of the proposed framework, ElasticNet, LightGBM, and XGBoost are 11.36%, 23.24%, 23.40%, and 22.09%, respectively. In terms of this metric, among the three feature-based methods, XGBoost leads to the minimum average MAPE of 22.09%, which is 1.94 times of 11.36% associated with the proposed framework. For the Zhu dataset, Figs. 7(c) and 7(d) show that among the three feature-based methods, LightGBM and XGBoost lead to the minimum errors in terms of the average RMSE and the average MAPE, respectively. For the Wang dataset, Figs. 7(e) and 7(f) show that among the three feature-based methods, ElasticNet results in the lowest errors in terms of both the average RMSE and the average MAPE. Although for the three datasets, the feature-based methods leading to the minimum errors in terms of different metrics vary, the proposed framework always leads to significantly lower errors even compared to these minimum errors. Therefore, these results show that the proposed framework is more effective than the three feature-based methods, which is mainly because it is rather difficult to extract effective features from a single cycle containing a very limited amount of information.

**Table 5**
Number and memory size of parameters of three feature-free models.

| Model | | AlexNet | ResNet18 | SqueezeNet |
|---|---|---|---|---|
| Number of parameters (millions) | Total | 61.10 | 11.18 | 0.72 |
| | Trainable | 61.10 | $5.13 \times 10^{-4}$ | $5.13 \times 10^{-4}$ |
| Memory size of parameters (MB) | | 233.08 | 42.64 | 2.76 |

**Table 6**
Model architecture and parameters of 1D-CNN.

| Layer index | Layer | Input size | Output size | Parameters (K, S, P) |
|---|---|---|---|---|
| 1 | Input | [B, 1, 1000] | – | – |
| 2 | Conv1d | [B, 1, 1000] | [B, 128, 1000] | 3 × 1, 1, 1 |
| 3 | BatchNorm1d | [B, 128, 1000] | – | – |
| 4 | ReLU | [B, 128, 1000] | – | – |
| 5 | MaxPool1d | [B, 128, 1000] | [B, 128, 500] | 2 × 1, 2, 0 |
| 6 | Conv1d | [B, 128, 500] | [B, 64, 500] | 3 × 1, 1, 1 |
| 7 | BatchNorm1d | [B, 64, 500] | – | – |
| 8 | ReLU | [B, 64, 500] | – | – |
| 9 | MaxPool1d | [B, 64, 500] | [B, 64, 250] | 2 × 1, 2, 0 |
| 10 | Conv1d | [B, 64, 250] | [B, 32, 250] | 3 × 1, 1, 1 |
| 11 | BatchNorm1d | [B, 32, 250] | – | – |
| 12 | ReLU | [B, 32, 250] | – | – |
| 13 | MaxPool1d | [B, 32, 250] | [B, 32, 125] | 2 × 1, 2, 0 |
| 14 | Flatten | [B, 32, 125] | [B, 4000] | – |
| 15 | Linear | [B, 4000] | [B, 100] | – |
| 16 | Dropout | [B, 100] | [B, 100] | $p = 0.2$ |
| 17 | Linear | [B, 100] | [B, 1] | – |

• B is the batch size, K is the kernel size, S is the stride, and P is the padding.
• $p$ is the dropout rate.

### 4.7. Evaluation results: Comparison with other feature-free methods

As another comparison, the proposed framework is evaluated against another three feature-free methods: ResNet18, SqueezeNet, and 1D-CNN. As mentioned in Section 4.1, all the layers of AlexNet are not frozen and they are trained directly based on the pretrained weights. For AlexNet, to reuse its results elaborated in Section 4.2, all the layers are still not frozen in this section. However, to investigate if ResNet18 and SqueezeNet can be implemented in a resource-constrained environment, all their layers are frozen except for the fully connected layer, which significantly reduces the number of the trainable parameters. The number and memory size of the parameters of AlexNet, ResNet18, and SqueezeNet are listed in Table 5. The model inputs of ResNet18 and SqueezeNet are the same as that of AlexNet. As for the 1D-CNN method, it is an end-to-end method and it utilizes the raw sequential data (i.e., the assembled capacity curve) as the model input. The model architecture and parameters of 1D-CNN are shown in Table 6. Again, all the three battery datasets are utilized in this evaluation. In the meantime, the assembled capacity curve for the second cycle is built using $Q_a = Q_c - Q_d$. For a fair comparison, the ResNet18, SqueezeNet, and 1D-CNN methods are also implemented using the setup illustrated in Section 4.1.

The comparison results are shown in Fig. 8, which lead to three observations. First, for the Severson dataset, Fig. 8(a) shows that the average RMSEs of AlexNet, ResNet18, SqueezeNet, and 1D-CNN are 121.02, 110.17, 135.61, and 180.33 cycles, respectively. Therefore, ResNet18 outperforms the other three models in terms of the average RMSE. Note that in this case, AlexNet still results in the second lowest RMSE, which is only higher than the one associated with ResNet18: 121.02 versus 110.17 cycles. On the other hand, Fig. 8(b) shows that the average MAPEs of AlexNet, ResNet18, SqueezeNet, and 1D-CNN are 11.36%, 11.69%, 13.09%, and 17.56%, respectively. Therefore, AlexNet performs the best among the four models in terms of the average MAPE. Second, for the Zhu and Wang datasets, AlexNet achieves the best performance in terms of both the average RMSE and the average MAPE. Finally, 1D-CNN performs the worst in all the three datasets in terms of both the average RMSE and the average MAPE. In general, AlexNet performs better compared to the other three feature-free methods, which again demonstrates the effectiveness of the proposed framework.

The evaluation results shown in Fig. 8 can be interpreted from a different perspective. In terms of the model input, AlexNet, ResNet18, and SqueezeNet are similar methods as their inputs are exactly the same while 1D-CNN is a method different from them. Clearly, 1D-CNN performs the worst among these four methods. Therefore, 1D-CNN may be of little value in real-world applications. Among the AlexNet, ResNet18, and SqueezeNet methods, AlexNet performs the best in general. Specifically, the performance of AlexNet is the best in Figs. 8(b) through 8(f) while its performance is only slightly poorer than ResNet18 in Fig. 8(a). Although the performance of ResNet18 and SqueezeNet is not as good as AlexNet, in general, their performance is at the same level as AlexNet. Note that compared to AlexNet, the slightly lower performance of ResNet18 and SqueezeNet is associated with relatively lower costs. As shown in Table 5, the numbers of the total parameters in ResNet18 and SqueezeNet are significantly lower than that in AlexNet: 11.18 and 0.72 versus 61.10 millions, respectively. In terms of percentages, ResNet18 and SqueezeNet only need 18.30% and 1.18% of the parameter number of AlexNet, respectively. In addition, the numbers of the trainable parameters in ResNet18 and SqueezeNet are both 513, which is negligible compared to 61.10 millions. Moreover, the memory sizes of the parameters in AlexNet, ResNet18, and SqueezeNet are 233.08, 42.64, and 2.76 MB, respectively. Therefore, it may be feasible to deploy the ultra-early battery cycle life prediction framework in a resource-constrained environment or a real-time battery monitoring application utilizing ResNet18 or SqueezeNet instead of AlexNet if a slight performance degradation is acceptable.

## 5. Conclusion

This paper proposes a battery cycle life prediction framework based on the assembled capacity curve extracted from a single cycle during the ultra-early stage of the battery cycling test. When the second cycle is utilized to build the assembled capacity curve, the average MAPEs are 11.36%, 15.27%, and 7.86% for the Severson, Zhu, and Wang datasets, respectively. Moreover, when the inverse assembled capacity curve is extracted from the second cycle, the prediction errors are approximately equal. In addition, when the assembled capacity curves are extracted from other early cycles (e.g., the fifth and tenth cycles), the prediction errors are also at the same level. Finally, the proposed framework outperforms three feature-based methods (i.e., ElasticNet, LightGBM, and XGBoost) and another three feature-free methods (i.e., ResNet18, SqueezeNet, and 1D-CNN). All these evaluation results demonstrate the effectiveness of the proposed framework in predicting the battery cycle life during an ultra-early stage.

While this paper conducts a rather comprehensive evaluation of the proposed framework, this paper can be further enhanced in various aspects. First, although the Severson, Zhu, and Wang datasets examined in this paper cover multiple commercially available batteries of different chemistries, manufacturers, models, and specifications, the three datasets are all about lithium-ion batteries. The applicability of the proposed framework to other chemistries such as lithium-metal batteries and sodium-ion batteries needs to be investigated. Different from lithium-ion batteries, publicly available datasets for such chemistries are limited. Therefore, in-house experiments need to be conducted. Second, all the three datasets examined in this paper are collected through well-designed experiments performed in the laboratory. However, for batteries equipped in real-world applications such as electric vehicles and energy storage stations, the SOH estimation and RUL prediction problems may be even more challenging. In fact, the authors have also considered such problems using the field data of electric

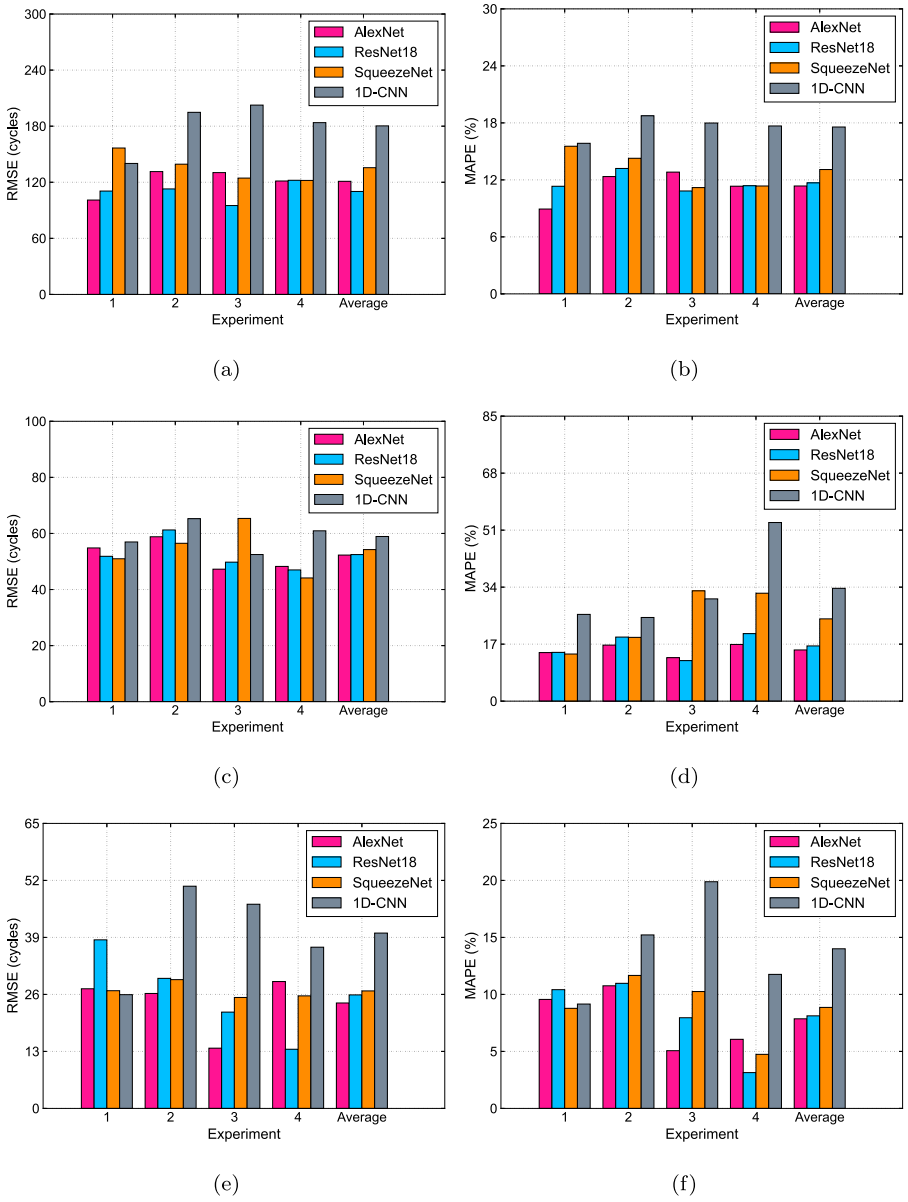



**Fig. 8.** Performance comparison between proposed framework and feature-free methods. (a) Severson dataset: RMSE. (b) Severson dataset: MAPE. (c) Zhu dataset: RMSE. (d) Zhu dataset: MAPE. (e) Wang dataset: RMSE. (f) Wang dataset: MAPE.

vehicles [31,32]. Such field data is significantly different from the laboratory data utilized in this paper. Therefore, the applicability of the proposed framework in such real-world scenarios also needs to be examined.

## CRediT authorship contribution statement

**Wenjin Yang:** Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Hengzhao Yang:** Conceptualization, Resources, Supervision, Project administration, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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